Facial Emotion Recognition

https://github.com/aiyshwary/Facial-Emotion-Detection

Python CNN SVM Random Forest TensorFlow Jupyter

OVERVIEW

A comparative study of machine learning and deep learning approaches for recognizing facial emotions from grayscale images, benchmarking CNN, SVM, Random Forest, and Decision Tree models across multiple datasets.

IMPACT

Implemented and benchmarked four ML/DL models for 7-class facial emotion classification. CNN achieved the best generalization at approximately 75.5% accuracy, outperforming all traditional models on unseen data.

TECHNICAL WALKTHROUGH

Project Explanation: Facial Emotion Recognition

This project focuses on facial emotion recognition using the icml_face_data dataset, which contains grayscale facial images labeled with seven emotion classes. The objective was to compare traditional machine learning models with a deep learning approach to identify the most effective method for emotion classification.

Initially, classical machine learning models such as Decision Tree, Random Forest, and Support Vector Machine (SVM) were trained using extracted pixel-based features. These models showed strong performance on the training dataset; however, significant overfitting was observed, particularly in Decision Tree and Random Forest models.

To address these limitations, a deep Convolutional Neural Network (CNN) was designed to automatically learn spatial and hierarchical features directly from facial images. The CNN architecture consisted of multiple convolutional blocks with batch normalization, max pooling, and dropout layers to improve generalization and reduce overfitting.

The models were evaluated using accuracy on the icml_face_data dataset and validation data to assess both performance and robustness.

One-line version

Although traditional models achieved high training accuracy, the CNN demonstrated superior generalization and robustness, making it the most suitable model for facial emotion recognition.

KEY DETAILS

1. Trained on icml_face_data and Kaggle FER datasets with 7 emotion classes (happy, sad, angry, fear, surprise, disgust, neutral).

2. CNN with data augmentation and stratified cross-validation achieved approximately 75.5% validation accuracy.

3. Random Forest reduced overfitting vs Decision Tree, achieving approximately 89.2% on ICML dataset.

4. SVM with PCA preprocessing achieved approximately 55% accuracy — lowest among all models on high-dimensional pixel data.

5. Concluded CNNs are best suited for FER due to hierarchical spatial feature extraction.

6. CNN architecture: 4 convolutional blocks (64-128-256-512 filters) with BatchNormalization, He Normal initialization, Dropout (0.25-0.6), and Dense(512) to Dense(7, softmax) head totaling ~7.1M parameters.

7. Input images are 48x48 grayscale pixels normalized to [0,1]; augmentation via Keras ImageDataGenerator with rotation (15 degrees), width/height shift, shear, zoom (0.15), and horizontal flip.

8. Training used Adam optimizer with EarlyStopping (patience=11), ReduceLROnPlateau (factor=0.5, patience=7, min_lr=1e-5), and 10-fold stratified cross-validation with batch size 32.